

KUBERNETES

Complete Commands Reference Guide

kubectl • Cluster • Pods • Services • Deployments • Troubleshooting

1. CLUSTER SETUP & INFO

kubectl Version & Config

Command	Description
<code>kubectl version</code>	Show client and server version
<code>kubectl version --client</code>	Show only client version
<code>kubectl cluster-info</code>	Show cluster endpoint info
<code>kubectl cluster-info dump</code>	Full cluster diagnostic dump
<code>kubectl config view</code>	Show kubeconfig file
<code>kubectl config get-contexts</code>	List all contexts
<code>kubectl config current-context</code>	Show current active context
<code>kubectl config use-context ollama-agent</code>	Switch to a different context
<code>kubectl config set-context --current --namespace=ollama-agent</code>	Set default namespace for current context
<code>kubectl api-resources</code>	List all API resource types
<code>kubectl api-versions</code>	List all API versions
<code>kubectl get nodes</code>	List all nodes in cluster
<code>kubectl get nodes -o wide</code>	List nodes with IP and OS info
<code>kubectl describe node ollama-agent-control-plane</code>	Detailed info about a node
<code>kubectl top nodes</code>	Show CPU/memory usage of nodes
<code>kubectl cordon <node></code>	Mark node as unschedulable
<code>kubectl uncordon <node></code>	Mark node as schedulable again
<code>kubectl drain <node> --ignore-daemonsets</code>	Evict all pods from a node

KIND Cluster Commands

Command	Description
<code>kind create cluster</code>	Create default KIND cluster
<code>kind create cluster --name ollama-agent</code>	Create named KIND cluster
<code>kind create cluster --name ollama-agent --config <file.yaml></code>	Create with custom config
<code>kind get clusters</code>	List all KIND clusters
<code>kind delete cluster --name ollama-agent</code>	Delete a KIND cluster

Command	Description
<code>kind load docker-image <image> --name <cluster></code>	Load local Docker image into KIND
<code>kind export kubeconfig --name ollama-agent</code>	Export kubeconfig for KIND cluster

2. NAMESPACES

Command	Description
<code>kubectl get namespaces</code>	List all namespaces
<code>kubectl get ns</code>	Short form — list namespaces
<code>kubectl create namespace ollama-agent</code>	Create a namespace
<code>kubectl delete namespace ollama-agent</code>	Delete namespace and ALL resources inside
<code>kubectl describe namespace ollama-agent</code>	Show namespace details and resource quotas
<code>kubectl get all -n ollama-agent</code>	List all resources in a namespace
<code>kubectl get all --all-namespaces</code>	List all resources across all namespaces
<code>kubectl get all -A</code>	Short form — all resources, all namespaces
<code>kubectl apply -f namespace.yaml</code>	Create namespace from YAML file
<code>-n ollama-agent</code>	Flag: target a specific namespace (used with any command)
<code>--namespace=ollama-agent</code>	Long form of -n flag
<code>--all-namespaces</code> or <code>-A</code>	Target all namespaces

[TIP] Always use `-n ollama-agent` with every `kubectl` command to avoid hitting the wrong namespace.

3. PODS

Get & Describe Pods

Command	Description
<code>kubectl get pods</code>	List pods in default namespace
<code>kubectl get pods -n ollama-agent</code>	List pods in specific namespace
<code>kubectl get pods -A</code>	List pods in all namespaces
<code>kubectl get pods -o wide</code>	List pods with node, IP, and more
<code>kubectl get pods -w</code>	Watch pods in real time
<code>kubectl get pods --show-labels</code>	Show pod labels
<code>kubectl get pods -l app=ollama-agent</code>	Filter pods by label
<code>kubectl describe pod ollama-backend-xxxxx -n ollama-agent</code>	Full details: events, status, containers
<code>kubectl get pod ollama-backend-xxxxx -o yaml</code>	Get pod spec as YAML
<code>kubectl get pod ollama-backend-xxxxx -o json</code>	Get pod spec as JSON

Pod Logs

Command	Description
<code>kubectl logs ollama-backend-xxxxx -n ollama-agent</code>	Get pod logs
<code>kubectl logs -f ollama-backend-xxxxx -n ollama-agent</code>	Stream logs in real time (-f = follow)
<code>kubectl logs ollama-backend-xxxxx -c <container> -n ollama-agent</code>	Logs from specific container in pod
<code>kubectl logs ollama-backend-xxxxx --previous -n ollama-agent</code>	Logs from crashed/previous container
<code>kubectl logs ollama-backend-xxxxx --tail=100 -n ollama-agent</code>	Last 100 lines of logs
<code>kubectl logs ollama-backend-xxxxx --since=1h -n ollama-agent</code>	Logs from last 1 hour
<code>kubectl logs -f deployment/ollama-agent -n ollama-agent</code>	Stream logs from a deployment
<code>kubectl logs -f statefulset/ollama-agent -n ollama-agent</code>	Stream logs from a statefulset

Pod Exec & Shell

Command	Description
<code>kubectl exec -it <pod> -n ollama-agent -- /bin/sh</code>	Open shell in pod (Alpine/slim images)
<code>kubectl exec -it <pod> -n ollama-agent -- /bin/bash</code>	Open bash shell in pod
<code>kubectl exec -it <pod> -c <container> -n ollama-agent -- /bin/sh</code>	Shell in specific container
<code>kubectl exec <pod> -n ollama-agent -- ls /app</code>	Run single command in pod
<code>kubectl exec <pod> -n ollama-agent -- env</code>	Print environment variables in pod
<code>kubectl exec <pod> -n ollama-agent -- cat /etc/hosts</code>	Print hosts file in pod
<code>kubectl exec -it statefulset/ollama-agent -n ollama-agent -- mongosh</code>	Open mongosh in MongoDB StatefulSet

Pod Lifecycle

Command	Description
<code>kubectl run ollama-agent --image=<image></code>	Create a standalone pod
<code>kubectl run ollama-agent --image=<image> --rm -it -- /bin/sh</code>	Temp debug pod — auto-deletes on exit
<code>kubectl delete pod ollama-backend-xxxxx -n ollama-agent</code>	Delete a pod (recreated if in Deployment)
<code>kubectl delete pod ollama-backend-xxxxx -n ollama-agent --force</code>	Force delete stuck pod
<code>kubectl delete pod ollama-backend-xxxxx -n ollama-agent --grace-period=0</code>	Immediate deletion (use carefully)
<code>kubectl get pod ollama-backend-xxxxx -n ollama-agent -o jsonpath='{.status.phase}'</code>	Get pod status programmatically

4. DEPLOYMENTS

Create & Manage Deployments

Command	Description
<code>kubectl get deployments -n ollama-agent</code>	List all deployments
<code>kubectl get deploy -n ollama-agent</code>	Short form
<code>kubectl describe deployment ollama-agent -n ollama-agent</code>	Full deployment details
<code>kubectl apply -f deployment.yaml</code>	Create/update deployment from file
<code>kubectl delete deployment ollama-agent -n ollama-agent</code>	Delete a deployment
<code>kubectl create deployment ollama-agent --image=<image></code>	Create deployment imperatively
<code>kubectl get deployment ollama-agent -o yaml -n ollama-agent</code>	Export deployment as YAML

Scaling & Rolling Updates

Command	Description
<code>kubectl scale deployment ollama-agent --replicas=3 -n ollama-agent</code>	Scale deployment to 3 replicas
<code>kubectl rollout status deployment/ollama-agent -n ollama-agent</code>	Watch rollout progress
<code>kubectl rollout history deployment/ollama-agent -n ollama-agent</code>	View rollout history
<code>kubectl rollout undo deployment/ollama-agent -n ollama-agent</code>	Rollback to previous version
<code>kubectl rollout undo deployment/ollama-agent --to-revision=2 -n ollama-agent</code>	Rollback to specific revision
<code>kubectl rollout restart deployment/ollama-agent -n ollama-agent</code>	Force restart all pods (rolling)
<code>kubectl rollout pause deployment/ollama-agent -n ollama-agent</code>	Pause a rolling update
<code>kubectl rollout resume deployment/ollama-agent -n ollama-agent</code>	Resume a paused rollout
<code>kubectl set image deployment/ollama-agent <container>=<image>:<tag> -n ollama-agent</code>	Update image in deployment

[OK] Always use `kubectl rollout status` after `kubectl apply` to confirm the rollout completed successfully.

5. SERVICES (SVC)

Get & Describe Services

Command	Description
<code>kubectl get services -n ollama-agent</code>	List all services

Command	Description
<code>kubectl get svc -n ollama-agent</code>	Short form
<code>kubectl get svc -A</code>	All services across all namespaces
<code>kubectl get svc -o wide -n ollama-agent</code>	Services with selector and ports
<code>kubectl describe svc ollama-agent -n ollama-agent</code>	Full service details with endpoints
<code>kubectl get endpoints -n ollama-agent</code>	List endpoints (pod IPs behind services)
<code>kubectl get ep ollama-agent -n ollama-agent</code>	Endpoints for specific service
<code>kubectl apply -f service.yaml</code>	Create/update service from file
<code>kubectl delete svc ollama-agent -n ollama-agent</code>	Delete a service
<code>kubectl get svc ollama-agent -o yaml -n ollama-agent</code>	Export service as YAML

Service Types Explained

Command	Description
<code>ClusterIP</code> (default)	Internal only — accessible inside cluster. No external access.
<code>NodePort</code>	Opens port (30000–32767) on every node. External access via <code>NodeIP:NodePort</code> .
<code>LoadBalancer</code>	Cloud load balancer (AWS ELB). Not available in KIND — use <code>port-forward</code> .
<code>ExternalName</code>	Maps service to external DNS name. No proxying.
<code>Headless</code> (<code>clusterIP: None</code>)	No load balancing. DNS returns individual pod IPs. Used by <code>StatefulSets</code> .

Port Forwarding

Command	Description
<code>kubectl port-forward svc/ollama-agent 8080:80 -n ollama-agent</code>	Forward local 8080 to service port 80
<code>kubectl port-forward pod/ollama-agent 8080:80 -n ollama-agent</code>	Forward to specific pod
<code>kubectl port-forward svc/ollama-agent 8080:80 -n ollama-agent --address 0.0.0.0</code>	Expose on all interfaces (EC2 access)
<code>kubectl port-forward svc/ollama-agent 8080:80 -n ollama-agent &</code>	Run port-forward in background
<code>kubectl port-forward -n ingress-nginx svc/ingress-nginx-controller 8080:80</code>	Forward Nginx Ingress controller

6. STATEFULSETS

Command	Description
<code>kubectl get statefulsets -n ollama-agent</code>	List all <code>StatefulSets</code>
<code>kubectl get sts -n ollama-agent</code>	Short form

Command	Description
<code>kubectl describe sts ollama-agent -n ollama-agent</code>	Detailed StatefulSet info
<code>kubectl apply -f statefulset.yaml</code>	Create/update StatefulSet from file
<code>kubectl delete sts ollama-agent -n ollama-agent</code>	Delete StatefulSet (pods also deleted)
<code>kubectl scale sts ollama-agent --replicas=3 -n ollama-agent</code>	Scale StatefulSet
<code>kubectl rollout status sts/ollama-agent -n ollama-agent</code>	Watch StatefulSet rollout
<code>kubectl rollout restart sts/ollama-agent -n ollama-agent</code>	Restart StatefulSet pods
<code>kubectl exec -it ollama-mongodb-0 -n ollama-agent -- /bin/sh</code>	Shell into first StatefulSet pod
<code>kubectl logs ollama-mongodb-0 -n ollama-agent</code>	Logs from first StatefulSet pod

[TIP] StatefulSet pods are named with stable index: mongodb-0, mongodb-1. Always start from index 0.

7. CONFIGMAPS & SECRETS

ConfigMaps

Command	Description
<code>kubectl get configmaps -n ollama-agent</code>	List ConfigMaps
<code>kubectl get cm -n ollama-agent</code>	Short form
<code>kubectl describe cm ollama-agent -n ollama-agent</code>	Show ConfigMap contents
<code>kubectl apply -f configmap.yaml</code>	Create/update ConfigMap from file
<code>kubectl delete cm ollama-agent -n ollama-agent</code>	Delete a ConfigMap
<code>kubectl create configmap ollama-agent --from-literal=KEY=VALUE -n ollama-agent</code>	Create ConfigMap from literal value
<code>kubectl create configmap ollama-agent --from-file=<file> -n ollama-agent</code>	Create ConfigMap from file
<code>kubectl get cm ollama-agent -o yaml -n ollama-agent</code>	Export ConfigMap as YAML

Secrets

Command	Description
<code>kubectl get secrets -n ollama-agent</code>	List all secrets
<code>kubectl describe secret ollama-agent -n ollama-agent</code>	Show secret metadata (not values)
<code>kubectl apply -f secret.yaml</code>	Create/update secret from file
<code>kubectl delete secret ollama-agent -n ollama-agent</code>	Delete a secret
<code>kubectl get secret ollama-agent -o yaml -n ollama-agent</code>	Export secret (values are base64)

Command	Description
<code>kubectl get secret ollama-agent -o jsonpath='{.data.KEY}' -n ollama-agent base64 --decode</code>	Decode a secret value
<code>kubectl create secret generic ollama-agent --from-literal=KEY=VALUE -n ollama-agent</code>	Create secret imperatively
<code>echo -n 'value' base64</code>	Encode a value for use in secret.yaml
<code>echo 'encoded==' base64 --decode</code>	Decode a base64 secret value

[WARN] Never commit secret.yaml with real values to Git. Always use .gitignore or external secret managers.

8. PERSISTENT VOLUMES & STORAGE

Command	Description
<code>kubectl get pvc -n ollama-agent</code>	List PersistentVolumeClaims
<code>kubectl get pv</code>	List PersistentVolumes (cluster-wide)
<code>kubectl describe pvc ollama-agent -n ollama-agent</code>	Full PVC details including status
<code>kubectl describe pv ollama-agent</code>	Full PV details
<code>kubectl get storageclass</code>	List storage classes
<code>kubectl describe storageclass ollama-agent</code>	Storage class details
<code>kubectl apply -f pvc.yaml</code>	Create PVC from file
<code>kubectl delete pvc ollama-agent -n ollama-agent</code>	Delete PVC (data may be lost!)
<code>kubectl get pvc -n ollama-agent -o wide</code>	PVC with storage class and capacity

[WARN] Deleting a PVC deletes the data inside it permanently. Always back up before deleting.

9. INGRESS

Command	Description
<code>kubectl get ingress -n ollama-agent</code>	List all Ingress resources
<code>kubectl get ing -n ollama-agent</code>	Short form
<code>kubectl get ingress -A</code>	Ingress across all namespaces
<code>kubectl describe ingress ollama-agent -n ollama-agent</code>	Full Ingress details with rules
<code>kubectl apply -f ingress.yaml</code>	Create/update Ingress from file
<code>kubectl delete ingress ollama-agent -n ollama-agent</code>	Delete an Ingress
<code>kubectl get pods -n ingress-nginx</code>	Check Nginx Ingress Controller pods
<code>kubectl logs -n ingress-nginx deploy/ingress-nginx-controller</code>	Ingress controller logs
<code>kubectl get svc -n ingress-nginx</code>	Check Ingress controller service

10. HPA — HORIZONTAL POD AUTOSCALER

Command	Description
<code>kubectl get hpa -n ollama-agent</code>	List all HPAs
<code>kubectl describe hpa ollama-agent -n ollama-agent</code>	HPA details: current vs target metrics
<code>kubectl apply -f hpa.yaml</code>	Create/update HPA from file
<code>kubectl delete hpa ollama-agent -n ollama-agent</code>	Delete an HPA
<code>kubectl autoscale deployment ollama-agent --min=2 --max=10 --cpu-percent=70 -n ollama-agent</code>	Create HPA imperatively
<code>kubectl top pods -n ollama-agent</code>	Show CPU/memory usage per pod (needs Metrics Server)
<code>kubectl top nodes</code>	Show CPU/memory usage per node

[TIP] HPA requires Metrics Server to be installed. Without it, HPA shows <unknown> for current metrics.

11. APPLY, DELETE & DRY RUN

Command	Description
<code>kubectl apply -f <file.yaml></code>	Create or update resource from file
<code>kubectl apply -f <directory>/</code>	Apply all YAML files in a directory
<code>kubectl apply -f <file.yaml> --dry-run=client</code>	Preview changes without applying
<code>kubectl apply -f <file.yaml> --dry-run=server</code>	Server-side dry run (validates with API)
<code>kubectl diff -f <file.yaml></code>	Show diff between live state and file
<code>kubectl delete -f <file.yaml></code>	Delete resource defined in file
<code>kubectl delete -f <directory>/</code>	Delete all resources in directory
<code>kubectl delete pod,svc,deploy --all -n ollama-agent</code>	Delete all pods, services, deployments in namespace
<code>kubectl create -f <file.yaml></code>	Create only (fails if already exists)
<code>kubectl replace -f <file.yaml></code>	Replace existing resource (must already exist)
<code>kubectl patch deployment ollama-agent -p '{...}' -n ollama-agent</code>	Patch a resource with JSON
<code>kubectl edit deployment ollama-agent -n ollama-agent</code>	Edit resource in-place with editor

12. LABELS & SELECTORS

Command	Description
<code>kubectl get pods --show-labels -n ollama-agent</code>	Show all labels on pods
<code>kubectl get pods -l app=backend -n ollama-agent</code>	Filter pods by label
<code>kubectl get pods -l app=backend,env=prod -n ollama-agent</code>	Filter by multiple labels

Command	Description
<code>kubectl label pod ollama-backend-xxxxx env=prod -n ollama-agent</code>	Add label to a pod
<code>kubectl label pod ollama-backend-xxxxx env- -n ollama-agent</code>	Remove label from a pod (note the dash)
<code>kubectl get all -l app=ollama-agent -n ollama-agent</code>	Get all resources with a label
<code>kubectl get pods -l 'app in (frontend,backend)' -n ollama-agent</code>	Filter by label set

13. EVENTS & MONITORING

Command	Description
<code>kubectl get events -n ollama-agent</code>	List all events in namespace
<code>kubectl get events -n ollama-agent --sort-by='.lastTimestamp'</code>	Events sorted by time (most recent last)
<code>kubectl get events -n ollama-agent --sort-by='.metadata.creationTimestamp'</code>	Events sorted by creation time
<code>kubectl get events -A</code>	Events across all namespaces
<code>kubectl get events -n ollama-agent --field-selector reason=Failed</code>	Filter only Failed events
<code>kubectl top pods -n ollama-agent</code>	CPU/memory per pod
<code>kubectl top pods -n ollama-agent --containers</code>	CPU/memory per container
<code>kubectl top nodes</code>	CPU/memory per node

14. TROUBLESHOOTING — A TO Z

Pod Not Starting / CrashLoopBackOff

[WARN] CrashLoopBackOff means the container keeps crashing. The `--previous` flag shows logs before the crash.

ImagePullBackOff / ErrImagePull

[OK] Causes: wrong image name, wrong tag, private registry with no `imagePullSecret`, typo in image name.

Pod Stuck in Pending

Service Not Routing Traffic

Ingress Not Working

PVC Stuck in Pending

HPA Showing <unknown> Metrics

StatefulSet Pod Not Starting

Node Not Ready

Force Delete Stuck Resources

[WARN] Force deleting pods and removing finalizers can cause data inconsistency. Use only when necessary.

15. FULL CLUSTER STATUS — QUICK CHECKS

One-liner full cluster status

16. OUTPUT FORMATS & JSONPATH

Command	Description
<code>kubectl get pods -o yaml -n ollama-agent</code>	Output as YAML
<code>kubectl get pods -o json -n ollama-agent</code>	Output as JSON
<code>kubectl get pods -o wide -n ollama-agent</code>	Wide output with extra columns
<code>kubectl get pods -o name -n ollama-agent</code>	Output only resource names
<code>kubectl get pod ollama-agent -o jsonpath='{.status.podIP}' -n ollama-agent</code>	Get pod IP address
<code>kubectl get pod ollama-agent -o jsonpath='{.spec.containers[*].name}' -n ollama-agent</code>	Get all container names
<code>kubectl get nodes -o jsonpath='{.items[*].metadata.name}'</code>	Get all node names
<code>kubectl get pods -n ollama-agent -l app=backend -o jsonpath='{.items[0].metadata.name}'</code>	Get first matching pod name
<code>kubectl get svc -n ollama-agent -o jsonpath='{.items[*].spec.clusterIP}'</code>	Get all ClusterIPs